# Skydrol® 5 Hydraulic Fluid

| Version | Revision Date: | SDS Number: | Date of last issue: - |
|---|---|---|---|
| 1.1 | 06/18/2019 | 150000093407 | Date of first issue: 09/06/2016 |
| PRD | | SDSUS / Z8 / 0001 | |

## SECTION 1. IDENTIFICATION

Product name : Skydrol® 5 Hydraulic Fluid

Product code : 34100-00, P3410005, P3410000, P3410004, P3410003, P3410001, P3410002, P3410006, P3410008, P3410009, P3410007, E3410001

**Manufacturer or supplier's details**

Company name of supplier : Eastman Chemical Company

Address : 200 South Wilcox Drive
Kingsport TN 37660-5280

Telephone : (423) 229-2000

Emergency telephone : CHEMTREC: +1-800-424-9300, +1-703-527-3887 CCN7321

**Recommended use of the chemical and restrictions on use**

Recommended use : Hydraulic fluids

Restrictions on use : None known.

## SECTION 2. HAZARDS IDENTIFICATION

**GHS classification in accordance with 29 CFR 1910.1200**

Skin sensitization : Category 1

**GHS label elements**

Hazard pictograms :

Signal Word : Warning

Hazard Statements : H317 May cause an allergic skin reaction.

Precautionary Statements :

**Prevention:**

P261 Avoid breathing dust/ fume/ gas/ mist/ vapors/ spray.
P272 Contaminated work clothing must not be allowed out of the workplace.
P280 Wear protective gloves.

**Response:**

P302 + P352 IF ON SKIN: Wash with plenty of soap and water.
P333 + P313 If skin irritation or rash occurs: Get medical advice/ attention.

P363 Wash contaminated clothing before reuse.

**Disposal:**

P501 Dispose of contents/ container to an approved waste disposal plant.

**Other hazards**

None known.

## SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS

**Components**

| Chemical name | CAS-No. | Concentration (% w/w) |
|---|---|---|
| Triisobutyl phosphate | 126-71-6 | 60 - 100 |
| 7-Oxabicyclo[4.1.0]heptane-3-carboxylic acid, 2-ethylhexyl ester | 62256-00-2 | 0 - 10 |
| tert-Butylphenyl diphenyl phosphate | 56803-37-3 | 1 - 5 |
| triphenylphosphate | 115-86-6 | 1 - 5 |
| butylated hydroxytoluene | 128-37-0 | 0.1 - 1 |

## SECTION 4. FIRST AID MEASURES

If inhaled : Move to fresh air.
If breathing is difficult, give oxygen.
Consult a physician if necessary.

In case of skin contact : Wash off immediately with plenty of water for at least 15 minutes.
Get medical attention if symptoms occur.
Wash contaminated clothing before reuse.

In case of eye contact : In case of contact, immediately flush eyes with plenty of water for at least 15 minutes.
Get medical attention if symptoms occur.

If swallowed : Call a physician or poison control center immediately.
Do not induce vomiting without medical advice.
Rinse mouth.
Never give anything by mouth to an unconscious person.

Most important symptoms and effects, both acute and delayed : May cause an allergic skin reaction.

Notes to physician : Treat symptomatically.

## SECTION 5. FIRE-FIGHTING MEASURES

Suitable extinguishing media : Water spray
Carbon dioxide (CO2)
Dry chemical

| | | |
|---|---|---|
| | | Foam |
| Unsuitable extinguishing media | : | Do not use a solid water stream as it may scatter and spread fire. |
| Hazardous combustion products | : | Carbon monoxide<br>Carbon dioxide (CO2)<br>oxides of phosphorus |
| Further information | : | Use a water spray to cool fully closed containers.<br>Do not allow run-off from fire fighting to enter drains or water courses. |
| Special protective equipment for fire-fighters | : | Wear an approved positive pressure self-contained breathing apparatus in addition to standard fire fighting gear. |

## SECTION 6. ACCIDENTAL RELEASE MEASURES

| | | |
|---|---|---|
| Personal precautions, protective equipment and emergency procedures | : | Ventilate the area.<br>Avoid breathing dust/ fume/ gas/ mist/ vapors/ spray.<br>Avoid contact with skin and eyes.<br>Material can create slippery conditions.<br>Wear appropriate personal protective equipment.<br>Local authorities should be advised if significant spillages cannot be contained. |
| Environmental precautions | : | Clear up spills immediately and dispose of waste safely.<br>Avoid release to the environment.<br>Collect spillage. |
| Methods and materials for containment and cleaning up | : | Contain spillage, soak up with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and transfer to a container for disposal according to local / national regulations (see section 13). |

## SECTION 7. HANDLING AND STORAGE

| | | |
|---|---|---|
| Advice on safe handling | : | Do not breathe vapors or spray mist.<br>Handle product only in closed system or provide appropriate exhaust ventilation at machinery.<br>In case of insufficient ventilation, wear suitable respiratory equipment.<br>Wear appropriate personal protective equipment.<br>Avoid contact with skin, eyes and clothing.<br>Wash thoroughly after handling.<br>Wash contaminated clothing before reuse.<br>Drain or remove substance from equipment prior to break-in or maintenance.<br>Handle in accordance with good industrial hygiene and safety practice. |
| Conditions for safe storage | : | Store locked up. |

Keep container tightly closed in a dry and well-ventilated place.
Keep in a cool place away from oxidizing agents.

## SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

**Ingredients with workplace control parameters**

| Components | CAS-No. | Value type (Form of exposure) | Control parameters / Permissible concentration | Basis |
|---|---|---|---|---|
| triphenylphosphate | 115-86-6 | TWA | 3 mg/m3 | ACGIH |
| | | TWA | 3 mg/m3 | NIOSH REL |
| | | TWA | 3 mg/m3 | OSHA Z-1 |
| | | TWA | 3 mg/m3 | OSHA P0 |

**Engineering measures** : Good general ventilation (typically 10 air changes per hour) should be used. Ventilation rates should be matched to conditions. If applicable, use process enclosures, local exhaust ventilation, or other engineering controls to maintain airborne levels below recommended exposure limits. If exposure limits have not been established, maintain airborne levels to an acceptable level.

**Personal protective equipment**

Respiratory protection : Use a properly fitted, particulate filter respirator complying with an approved standard if a risk assessment indicates this is necessary.
Respirator selection, use, and maintenance must be in accordance with regulatory requirements, if applicable.
If engineering controls do not maintain airborne concentrations below recommended exposure limits (where applicable) or to an acceptable level (in countries where exposure limits have not been established), an approved respirator must be worn.

Hand protection

Remarks : Wear suitable gloves. Contact the glove manufacturer for specific advice on glove selection and breakthrough times for your use conditions. After contamination with product change the gloves immediately and dispose of them according to relevant national and local regulations.

Eye protection : Wear safety glasses with side shields (or goggles).

Skin and body protection : Wear suitable protective clothing.

Protective measures : Ensure that eye flushing systems and safety showers are located close to the working place.

## SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

| | | |
|---|---|---|
| Appearance | : | oily |
| Color | : | purple |
| Odor | : | odorless |
| Odor Threshold | : | not determined |
| pH | : | not determined |
| Melting point/freezing point | : | < -80 °F / < -62 °C |
| Boiling point/boiling range | : | not determined |
| Flash point | : | 309 °F / 154 °C<br>Method: Cleveland open cup |
| Evaporation rate | : | not determined |
| Self-ignition | : | > 750 °F / > 399 °C<br>Method: ASTM D2155 |
| Upper explosion limit / Upper flammability limit | : | not determined |
| Lower explosion limit / Lower flammability limit | : | not determined |
| Vapor pressure | : | 0.4 hPa (77 °F / 25 °C) |
| Relative vapor density | : | not determined |
| Relative density | : | 0.970 - 0.980 (77 °F / 25 °C) |
| Density | : | 977 kg/m3 (77 °F / 25 °C) |
| Solubility(ies) | | |
| Water solubility | : | slightly soluble |
| Partition coefficient: n-octanol/water | : | No data available |
| Autoignition temperature | : | not determined |
| Decomposition temperature | : | not determined |
| Viscosity | | |
| Viscosity, dynamic | : | not determined |
| Viscosity, kinematic | : | < 2600 mm2/s (-65 °F / -54 °C) |

| | | |
|---|---|---|
| | | 9.00 - 9.84 mm2/s (100 °F / 38 °C) |
| | | 3.01 - 3.41 mm2/s (210 °F / 99 °C) |
| Explosive properties | : | Not classified |
| Oxidizing properties | : | Not classified |

## SECTION 10. STABILITY AND REACTIVITY

| | | |
|---|---|---|
| Reactivity | : | None reasonably foreseeable. |
| Chemical stability | : | Stable under normal conditions. |
| Possibility of hazardous reactions | : | None known. |
| Conditions to avoid | : | None known. |
| Incompatible materials | : | Strong oxidizing agents |
| Hazardous decomposition products | : | Emits acrid smoke and fumes when heated to decomposition. |

## SECTION 11. TOXICOLOGICAL INFORMATION

**Acute toxicity**

Not classified based on available information.

**Product:**

| | | |
|---|---|---|
| Acute oral toxicity | : | Remarks: No data available |
| Acute inhalation toxicity | : | Remarks: No data available |
| Acute dermal toxicity | : | Remarks: No data available |

**Components:**

**Triisobutyl phosphate:**

| | | |
|---|---|---|
| Acute oral toxicity | : | LD50 Oral (Rat): > 5,000 mg/kg<br>Assessment: The substance or mixture has no acute oral toxicity |
| Acute inhalation toxicity | : | LC50 (Rat): > 5.14 mg/l<br>Exposure time: 4 h<br>Test atmosphere: dust/mist<br>Assessment: The substance or mixture has no acute inhalation toxicity |
| Acute dermal toxicity | : | LD50 Dermal (Rabbit): > 5,000 mg/kg<br>Assessment: The substance or mixture has no acute dermal toxicity |

**7-Oxabicyclo[4.1.0]heptane-3-carboxylic acid, 2-ethylhexyl ester:**

Acute dermal toxicity : LD50 Dermal (Rabbit, Male and Female): > 7,940 mg/kg

**tert-Butylphenyl diphenyl phosphate:**

Acute oral toxicity : LDLo (Rat, Male and Female): 15,800 mg/kg
Assessment: Not classified

Acute inhalation toxicity : (Rat): > 0.4 mg/l

Acute dermal toxicity : LDLo (Rabbit, Male and Female): > 7,940 mg/kg
Assessment: Not classified

**triphenylphosphate:**

Acute oral toxicity : LD50 Oral (Rat, male): > 6,400 mg/kg

Acute dermal toxicity : LD50 Dermal (Guinea pig, male): > 5,000 mg/kg

**Skin corrosion/irritation**

Not classified based on available information.

**Product:**

Remarks : No data available

**Components:**

**Triisobutyl phosphate:**

Species : Rabbit
Exposure time : 4 h
Result : Mild skin irritation

**7-Oxabicyclo[4.1.0]heptane-3-carboxylic acid, 2-ethylhexyl ester:**

Species : Rabbit
Exposure time : 24 h
Assessment : Not classified as hazardous.
Result : slight to moderate irritation

**tert-Butylphenyl diphenyl phosphate:**

Species : Rabbit
Assessment : Not classified
Method : Draize Test
Result : slight irritation

**triphenylphosphate:**

Species : Guinea pig
Remarks : Non-irritating to the skin.

**Serious eye damage/eye irritation**

Not classified based on available information.

**Product:**

Remarks : No data available

**Components:**

**Triisobutyl phosphate:**

Species : Rabbit
Result : slight
Assessment : Not classified

**7-Oxabicyclo[4.1.0]heptane-3-carboxylic acid, 2-ethylhexyl ester:**

Species : Rabbit
Result : slight irritation
Exposure time : 24 h
Assessment : Not classified

**tert-Butylphenyl diphenyl phosphate:**

Species : Rabbit
Result : slight irritation
Assessment : Not classified

**triphenylphosphate:**

Species : Rabbit
Result : slight

**Respiratory or skin sensitization**

**Skin sensitization**

May cause an allergic skin reaction.

**Respiratory sensitization**

Not classified based on available information.

**Components:**

**Triisobutyl phosphate:**

Test Type : OECD 406: Guinea pig sensitization
Species : Guinea pig
Method : OECD 406: Guinea pig sensitization
Result : May cause sensitization by skin contact.

**7-Oxabicyclo[4.1.0]heptane-3-carboxylic acid, 2-ethylhexyl ester:**

Test Type : Skin Sensitization
Species : Guinea pig
Result : May cause sensitization by skin contact.

**tert-Butylphenyl diphenyl phosphate:**

| | | |
|---|---|---|
| Test Type | : | OECD 429: LLNA |
| Species | : | Mouse |
| Assessment | : | Not classified |
| Result | : | sensitizing |
| | | |
| Species | : | Humans |
| Assessment | : | Not classified |
| Result | : | Not a skin sensitizer. |
| Remarks | : | Patch test on human volunteers did not demonstrate sensitization properties. |

**Germ cell mutagenicity**

Not classified based on available information.

**Components:**

**Triisobutyl phosphate:**

| | | |
|---|---|---|
| Genotoxicity in vitro | : | Test Type: Salmonella typhimurium assay (Ames test)<br>Metabolic activation: +/- activation<br>Method: Bacterial Reverse Mutation Assay<br>Result: negative |
| Genotoxicity in vivo | : | Result: negative |

**7-Oxabicyclo[4.1.0]heptane-3-carboxylic acid, 2-ethylhexyl ester:**

| | | |
|---|---|---|
| Genotoxicity in vitro | : | Test Type: Salmonella typhimurium assay (Ames test)<br>Metabolic activation: +/- activation<br>Method: Bacterial Reverse Mutation Assay<br>Result: negative<br><br>Test Type: Mutagenicity - Mammalian<br>Metabolic activation: +/- activation<br>Method: In vitro Mammalian Chromosome Aberration Test<br>Result: equivocal<br><br>Test Type: Mutagenicity - Mammalian<br>Metabolic activation: +/- activation<br>Method: In vitro Mammalian Cell Gene Mutation Test<br>Result: negative |
| Genotoxicity in vivo | : | Species: Rat (Male and Female)<br>Application Route: intraperitoneal injection<br>Method: Mammalian Bone Marrow Chromosome Aberration Test<br>Result: equivocal |

**tert-Butylphenyl diphenyl phosphate:**

| | | |
|---|---|---|
| Genotoxicity in vitro | : | Test Type: Mutagenicity - Bacterial<br>Method: Bacterial Reverse Mutation Assay<br>Result: negative |

Test Type: Mutagenicity - Mammalian
Method: In vitro Mammalian Cell Gene Mutation Test
Result: negative

Test Type: Mutagenicity - Mammalian
Method: Genetic Toxicology: In Vitro Sister Chromatid Exchange Assay in Mammalian Cells
Result: negative

**Carcinogenicity**

Not classified based on available information.

**IARC** No ingredient of this product present at levels greater than or equal to 0.1% is identified as probable, possible or confirmed human carcinogen by IARC.

**OSHA** No component of this product present at levels greater than or equal to 0.1% is on OSHA's list of regulated carcinogens.

**NTP** No ingredient of this product present at levels greater than or equal to 0.1% is identified as a known or anticipated carcinogen by NTP.

**Reproductive toxicity**

Not classified based on available information.

**Product:**

Effects on fertility : Remarks: No data available

**Components:**

**triphenylphosphate:**

Reproductive toxicity - Assessment : Based on available data, the classification criteria are not met.

**STOT-single exposure**

Not classified based on available information.

**Product:**

Routes of exposure : inhalation (dust/mist/fume)
Assessment : Not classified

**Components:**

**Triisobutyl phosphate:**

Assessment : Not classified

**STOT-repeated exposure**

Not classified based on available information.

**Product:**

Assessment : Not classified

**Components:**

**Triisobutyl phosphate:**

Assessment : Not classified

**tert-Butylphenyl diphenyl phosphate:**

Routes of exposure : Oral
Assessment : Based on available data, the classification criteria are not met.

**Repeated dose toxicity**

**Components:**

**Triisobutyl phosphate:**

Species : Rat, male
: 68.4 mg/kg
Application Route : Oral Study
Exposure time : 90 days

**tert-Butylphenyl diphenyl phosphate:**

Species : Rat, Male and Female
NOAEL : 107.5 - 124.8 mg/kg
Application Route : Oral Study
Exposure time : 90 days
Remarks : Based on available data, the classification criteria are not met.

**Aspiration toxicity**

Not classified based on available information.

**Product:**

No data available

**Components:**

**Triisobutyl phosphate:**

Not classified

**7-Oxabicyclo[4.1.0]heptane-3-carboxylic acid, 2-ethylhexyl ester:**

Not applicable

**Information on likely routes of exposure**

**Product:**

Inhalation : Remarks: None known.

Skin contact : Remarks: May cause an allergic skin reaction.

Eye contact : Remarks: Contact with the eyes may be very painful but does not cause damage.

Ingestion : Remarks: None known.

## SECTION 12. ECOLOGICAL INFORMATION

### Ecotoxicity

**Product:**

Toxicity to fish : LC50 (Oncorhynchus mykiss (rainbow trout)): 23 mg/l
Exposure time: 96 h
Remarks: Read-across from a similar material

EC50 (Leuciscus idus (Golden orfe)): 17.8 - 21.5 mg/l
Exposure time: 96 h
Remarks: Read-across from a similar material

Toxicity to daphnia and other aquatic invertebrates : (Daphnia magna (Water flea)): 50.5 mg/l
Exposure time: 48 h
Remarks: Read-across from a similar material

EC50 (Daphnia magna (Water flea)): 11 mg/l
Exposure time: 48 h

Toxicity to algae/aquatic plants : ErC50 (Scenedesmus subspicatus): 34.1 mg/l
Exposure time: 72 h

Toxicity to fish (Chronic toxicity) : NOEC:
Remarks: No data available

Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity) : NOEC:
Remarks: No data available

Toxicity to microorganisms : EC50 (Bacteria): 37.2 mg/l
Exposure time: 28 d

**Components:**

**Triisobutyl phosphate:**

Toxicity to fish : EC50 (Leuciscus idus (Golden orfe)): 17.8 - 21.5 mg/l
Exposure time: 96 h

Toxicity to daphnia and other aquatic invertebrates : EC50 (Daphnia magna (Water flea)): 11 mg/l
Exposure time: 48 h

Toxicity to algae/aquatic plants : ErC50 (Scenedesmus subspicatus): 34.1 mg/l
Exposure time: 72 h

Toxicity to microorganisms : EC50 (Bacteria): 37.2 mg/l
Exposure time: 28 d

**tert-Butylphenyl diphenyl phosphate:**

| | | |
|---|---|---|
| Toxicity to fish | : | EC50 (Ictalurus catus (catfish)): 0.8 mg/l<br>Exposure time: 96 h |
| Toxicity to daphnia and other aquatic invertebrates | : | EC50 (Daphnia magna (Water flea)): 0.2 mg/l<br>Exposure time: 48 h |
| M-Factor (Acute aquatic toxicity) | : | 1 |

**triphenylphosphate:**

| | | |
|---|---|---|
| Toxicity to fish | : | LC50 (Oncorhynchus mykiss (rainbow trout)): 0.4 mg/l<br>Exposure time: 96 h |
| M-Factor (Acute aquatic toxicity) | : | 1 |

**Persistence and degradability**

**Product:**

| | | |
|---|---|---|
| Biodegradability | : | Remarks: Expected to be biodegradable |
| Biochemical Oxygen Demand (BOD) | : | Remarks: No data available |
| Chemical Oxygen Demand (COD) | : | Remarks: No data available |

**Components:**

**Triisobutyl phosphate:**

| | | |
|---|---|---|
| Biodegradability | : | Result: Readily biodegradable.<br>Biodegradation: 70 - 80 %<br>Exposure time: 28 d<br>Method: Ready Biodegradability: CO2 Evolution Test |

**7-Oxabicyclo[4.1.0]heptane-3-carboxylic acid, 2-ethylhexyl ester:**

| | | |
|---|---|---|
| Biodegradability | : | Concentration: 100 mg/l<br>Method: Ready Biodegradability: Modified MITI Test (I)<br>Remarks: Readily biodegradable |

**tert-Butylphenyl diphenyl phosphate:**

| | | |
|---|---|---|
| Biodegradability | : | Result: Readily biodegradable. |

**triphenylphosphate:**

| | | |
|---|---|---|
| Biodegradability | : | Result: Readily biodegradable. |

**Bioaccumulative potential**

**Product:**

Bioaccumulation : Remarks: Bioaccumulation is unlikely.
Read-across from a similar material

**Components:**

**Triisobutyl phosphate:**

Bioaccumulation : Remarks: Bioaccumulation is unlikely.

Partition coefficient: n-octanol/water : log Pow: 3.72

**tert-Butylphenyl diphenyl phosphate:**

Bioaccumulation : Species: Fish
Bioconcentration factor (BCF): 1,850

**triphenylphosphate:**

Partition coefficient: n-octanol/water : log Pow: 4.63

**Mobility in soil**

No data available

**Other adverse effects**

No data available

## SECTION 13. DISPOSAL CONSIDERATIONS

**Disposal methods**

Waste from residues : This product meets the criteria for a synthetic used oil under the U.S. EPA Standards for the Management of Used Oil (40 CFR 279). Those standards govern recycling and disposal in lieu of 40 CFR 260 -272 of the Federal hazardous waste program in states that have adopted these used oil regulations. Consult your attorney or appropriate regulatory official to be sure these standards have been adopted in your state. Recycle or burn in accordance with the applicable standards.
Dispose of in accordance with local regulations.

## SECTION 14. TRANSPORT INFORMATION

**International Regulations**

**IATA-DGR**

UN/ID No. : UN 3082
Proper shipping name : Environmentally hazardous substance, liquid, n.o.s. (triphenyl phosphate)

| | | |
|---|---|---|
| Class | : | 9 |
| Packing group | : | III |
| Labels | : | Miscellaneous |
| Packing instruction (cargo aircraft) | : | 964 |
| Packing instruction (passenger aircraft) | : | 964 |
| Remarks | : | Shipping in package sizes of less than 5 L (liquids) or 5 KG (solids) may lead to a non-regulated classification. |

**IMDG-Code**

| | | |
|---|---|---|
| UN number | : | UN 3082 |
| Proper shipping name | : | ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (triphenyl phosphate) |
| Class | : | 9 |
| Packing group | : | III |
| Labels | : | 9 |
| EmS Code | : | F-A, S-F |
| Marine pollutant | : | yes |
| Remarks | : | Shipping in package sizes of less than 5 L (liquids) or 5 KG (solids) may lead to a non-regulated classification. |

**Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code**

Not applicable for product as supplied.

**Domestic regulation**

**49 CFR**

| | | |
|---|---|---|
| UN/ID/NA number | : | UN 3082 |
| Proper shipping name | : | Environmentally hazardous substance, liquid, n.o.s. (triphenyl phosphate) |
| Class | : | 9 |
| Packing group | : | III |
| Labels | : | Class 9 - Miscellaneous Dangerous Goods |
| ERG Code | : | 171 |
| Marine pollutant | : | yes(triphenyl phosphate) |
| Remarks | : | Above applies only to containers over 119 gallons or 450 liters. Not regulated if shipped in packages less than or equal to 119 gallons (450 liters). |

**Special precautions for user**

The transport classification(s) provided herein are for informational purposes only, and solely based upon the properties of the unpackaged material as it is described within this Safety Data Sheet. Transportation classifications may vary by mode of transportation, package sizes, and variations in regional or country regulations.

## SECTION 15. REGULATORY INFORMATION

**EPCRA - Emergency Planning and Community Right-to-Know**

**CERCLA Reportable Quantity**

This material does not contain any components with a CERCLA RQ.

**SARA 304 Extremely Hazardous Substances Reportable Quantity**

This material does not contain any components with a section 304 EHS RQ.

**SARA 302 Extremely Hazardous Substances Threshold Planning Quantity**

This material does not contain any components with a section 302 EHS TPQ.

| | | |
|---|---|---|
| **SARA 311/312 Hazards** | : | Respiratory or skin sensitization |
| **SARA 313** | : | This material does not contain any chemical components with known CAS numbers that exceed the threshold (De Minimis) reporting levels established by SARA Title III, Section 313. |

**California Prop. 65**

This product does not contain any chemicals known to the State of California to cause cancer, birth, or any other reproductive defects.

**The ingredients of this product are reported in the following inventories:**

| | | |
|---|---|---|
| DSL | : | On the inventory, or in compliance with the inventory |
| AICS | : | On the inventory, or in compliance with the inventory |
| ENCS | : | On the inventory, or in compliance with the inventory |
| KECI | : | Not listed |
| PICCS | : | Not listed |
| IECSC | : | On the inventory, or in compliance with the inventory |
| TSCA | : | On the inventory, or in compliance with the inventory |

**TSCA list**

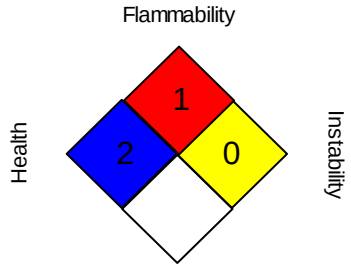
No substances are subject to a Significant New Use Rule.

No substances are subject to TSCA 12(b) export notification requirements.

## SECTION 16. OTHER INFORMATION

### Further information

**NFPA 704:**




Special hazard.

**HMIS® IV:**

| | | |
|---|---|---|
| HEALTH | / | 2 |
| FLAMMABILITY | | 1 |
| PHYSICAL HAZARD | | 0 |

HMIS® ratings are based on a 0-4 rating scale, with 0 representing minimal hazards or risks, and 4 representing significant hazards or risks. The "*" represents a chronic hazard, while the "/" represents the absence of a chronic hazard.

### Full text of other abbreviations

| | | |
|---|---|---|
| ACGIH | : | USA. ACGIH Threshold Limit Values (TLV) |
| NIOSH REL | : | USA. NIOSH Recommended Exposure Limits |
| OSHA P0 | : | USA. OSHA - TABLE Z-1 Limits for Air Contaminants - 1910.1000 |
| OSHA Z-1 | : | USA. Occupational Exposure Limits (OSHA) - Table Z-1 Limits for Air Contaminants |
| ACGIH / TWA | : | 8-hour, time-weighted average |
| NIOSH REL / TWA | : | Time-weighted average concentration for up to a 10-hour workday during a 40-hour workweek |
| OSHA P0 / TWA | : | 8-hour time weighted average |
| OSHA Z-1 / TWA | : | 8-hour time weighted average |

AICS - Australian Inventory of Chemical Substances; ASTM - American Society for the Testing of Materials; bw - Body weight; CERCLA - Comprehensive Environmental Response, Compensation, and Liability Act; CMR - Carcinogen, Mutagen or Reproductive Toxicant; DIN - Standard of the German Institute for Standardisation; DOT - Department of Transportation; DSL - Domestic Substances List (Canada); ECx - Concentration associated with x% response; EHS - Extremely Hazardous Substance; ELx - Loading rate associated with x% response; EmS - Emergency Schedule; ENCS - Existing and New Chemical Substances (Japan); ErCx - Concentration associated with x% growth rate response; ERG - Emergency Response Guide; GHS - Globally Harmonized System; GLP - Good Laboratory Practice; HMIS - Hazardous Materials Identification System; IARC - International Agency for Research on Cancer; IATA - International Air Transport Association; IBC - International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals in Bulk; IC50 - Half maximal inhibitory concentration; ICAO - International Civil Aviation Organization; IECSC - Inventory of Existing Chemical Substances in China; IMDG - International Maritime Dangerous Goods; IMO - International Maritime Organization; ISHL - Industrial Safety and Health Law (Japan); ISO - International Organisation for Standardization; KECI - Korea Existing Chemicals Inventory; LC50 - Lethal Concentration to 50 % of a test population; LD50 - Lethal Dose to 50% of a test population (Median Lethal Dose); MARPOL - International Convention for the Prevention of Pollution from Ships; MSHA - Mine Safety and Health Administration; n.o.s. - Not Otherwise Specified; NFPA - National Fire Protection Association; NO(A)EC - No Observed (Adverse)

Effect Concentration; NO(A)EL - No Observed (Adverse) Effect Level; NOELR - No Observable Effect Loading Rate; NTP - National Toxicology Program; NZIoC - New Zealand Inventory of Chemicals; OECD - Organization for Economic Co-operation and Development; OPPTS - Office of Chemical Safety and Pollution Prevention; PBT - Persistent, Bioaccumulative and Toxic substance; PICCS - Philippines Inventory of Chemicals and Chemical Substances; (Q)SAR - (Quantitative) Structure Activity Relationship; RCRA - Resource Conservation and Recovery Act; REACH - Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals; RQ - Reportable Quantity; SADT - Self-Accelerating Decomposition Temperature; SARA - Superfund Amendments and Reauthorization Act; SDS - Safety Data Sheet; TCSI - Taiwan Chemical Substance Inventory; TSCA - Toxic Substances Control Act (United States); UN - United Nations; UNRTDG - United Nations Recommendations on the Transport of Dangerous Goods; vPvB - Very Persistent and Very Bioaccumulative

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| Sources of key data used to compile the Material Safety Data Sheet | : | www.EastmanAviationSolutions.com |
| Revision Date | : | 06/18/2019 |

The information provided in this Material Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

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